

परीक्षक-AI

PARIKSHAK-AI: an AI copilot for on-screen marking

The AI pre-reads every handwritten answer, in Hindi or English. The examiner decides every mark.

Team eMitra

github.com/nishchaydev/Mphack

THE PROBLEM

Evaluation in Madhya Pradesh is breaking down

~30%

of BU Bhopal results from an outsourced agency suspected wrong; nearly 10,000 complaints

43%

B.Ed pass rate at DAVV Indore before an expert re-evaluation of answer sheets was ordered

3–14

months of result delay across MP state universities, against a 30–45 day norm

Root cause: 3.04 Crore answer scripts a year, examiners paid ₹16–25 a booklet, 36-page scripts glance-checked in minutes, and most general-stream students write in Hindi.

WHO IT HURTS

Four stakeholders, one broken system

The examiner

₹16–25 a booklet, 40 scripts a sitting, no rubric help.

→ An AI pre-read with quoted evidence; the final mark stays theirs.

Controller of Examinations

No live view; 5% sampling after results.

→ Live dashboard, blind seed scripts, one-click audit dossiers.

The student

Months of delay; ₹500–1,500 per revaluation.

→ Faster, checked results and feedback on each answer.

MPOnline & Higher Education

Fragmented, risky vendor tenders.

→ A bolt-on layer on the existing portal; no migration.

OUR APPROACH

We don't replace the examiner. We give them an AI copilot, and give the university proof that every script was read.

Human in the loop

The AI suggests a mark per criterion with the sentence that earns it. The examiner accepts, modifies or overrides.

Bolt-on, not replace

Plugs into MPOnline's ASP.NET / Oracle portal through pre-signed URLs, APIs and a queue.

Sovereign by design

In production, data and inference stay in the MP State Data Centre or an India-region cloud.

ARCHITECTURE

A bolt-on layer; MPOnline's systems stay untouched

EXISTING

MPOnline portal
ASP.NET · Oracle 19c ·
State Data Centre

University scanning
Non-destructive, 300 DPI



OUTPUTS

Examiner workspace
Accept, modify or override

Tabulation register
Final marks, batch sync

CoE command centre
Live integrity view

END TO END

From answer booklet to published result



Target: results in 15–20 days instead of months.

Read Hindi directly, and calibrate examiners blind

1 · HANDWRITING RECOGNITION ASSISTANCE

Reads Hindi handwriting directly

Old: image → OCR → broken Devanagari → an LLM guesses.

Ours: image + rubric → vision model → a mark per criterion with the student's own words as evidence.

“केशवानंद भारती केस (1973) में मूल ढाँचे का सिद्धांत दिया गया”

→ Indian-context criterion: 2 / 2, quote shown to the examiner

2 · SMART MODERATION

Blind seed scripts

- The Chief Examiner pre-marks anchor scripts
- They sit in examiners' queues and look like any other script
- Drift above 15% of max marks: CoE alert, and the next scripts go to the Head Examiner
- The AI's own suggestions are checked against the same anchors

Standard practice in Cambridge and IB online marking.

Catch rushed marking and organised copying

3 · MARKING ANOMALIES

Reading-time check, never a lockout

$$T_{\min} = (\text{words} \div 200 + 0.5 \times \text{equations} + 0.75 \times \text{diagrams}) \\ \min + 10 \text{ s}$$

- Under 40% of the floor, or a page unopened: a soft prompt
- Under 25%: check one criterion or mark the page to continue
- Repeated rushing: a silent second reading

4 · MALPRACTICE DETECTION

Exam-centre collusion audit

$$\text{RPI} = (\text{similarity within centre} - \text{state mean}) \div \text{state std. dev.}$$

- Compares answers within each exam centre against the statewide baseline
- Flags clusters sharing the same unusual arguments or errors
- Offline audit before results are published (planned)

Transparent results and fairer marking

5 · FASTER, TRANSPARENT RESULTS

One-click audit dossier

- Annotated scans and the evidence behind every mark
- Time spent on each page
- SHA-256 Merkle root, signed, on a hash-chained ledger
- Any later edit to stored marks is detected
- Supplies the particulars for a Section 63 BSA certificate

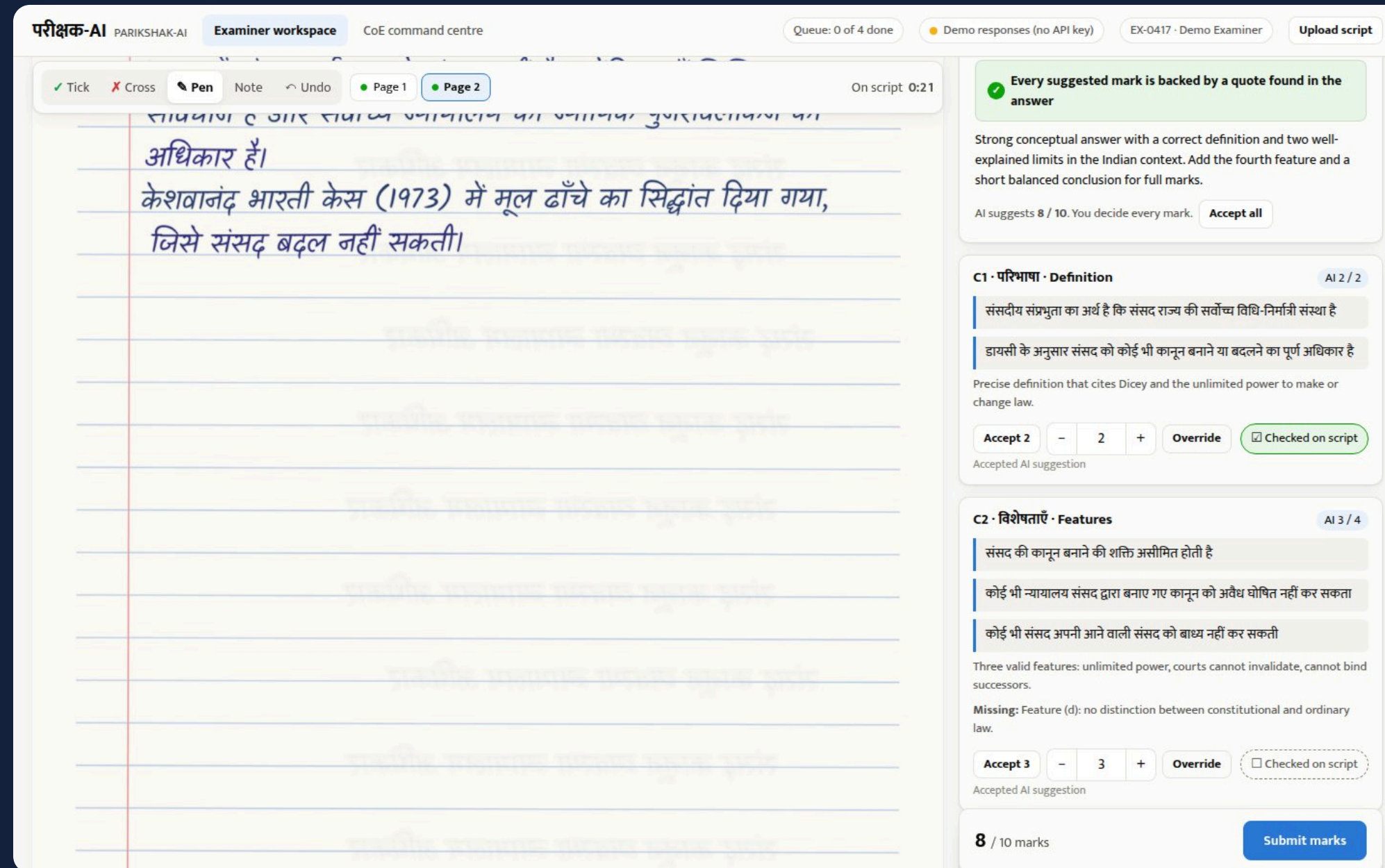
6 · EXAMINER ANALYTICS

Item-level marking

- Booklets split into question clips
- One examiner marks one question across many scripts
- Reduces the halo effect; faculty mark their strongest topics
- Used by Cambridge, IB and Pearson (planned)

WORKING PROOF OF CONCEPT

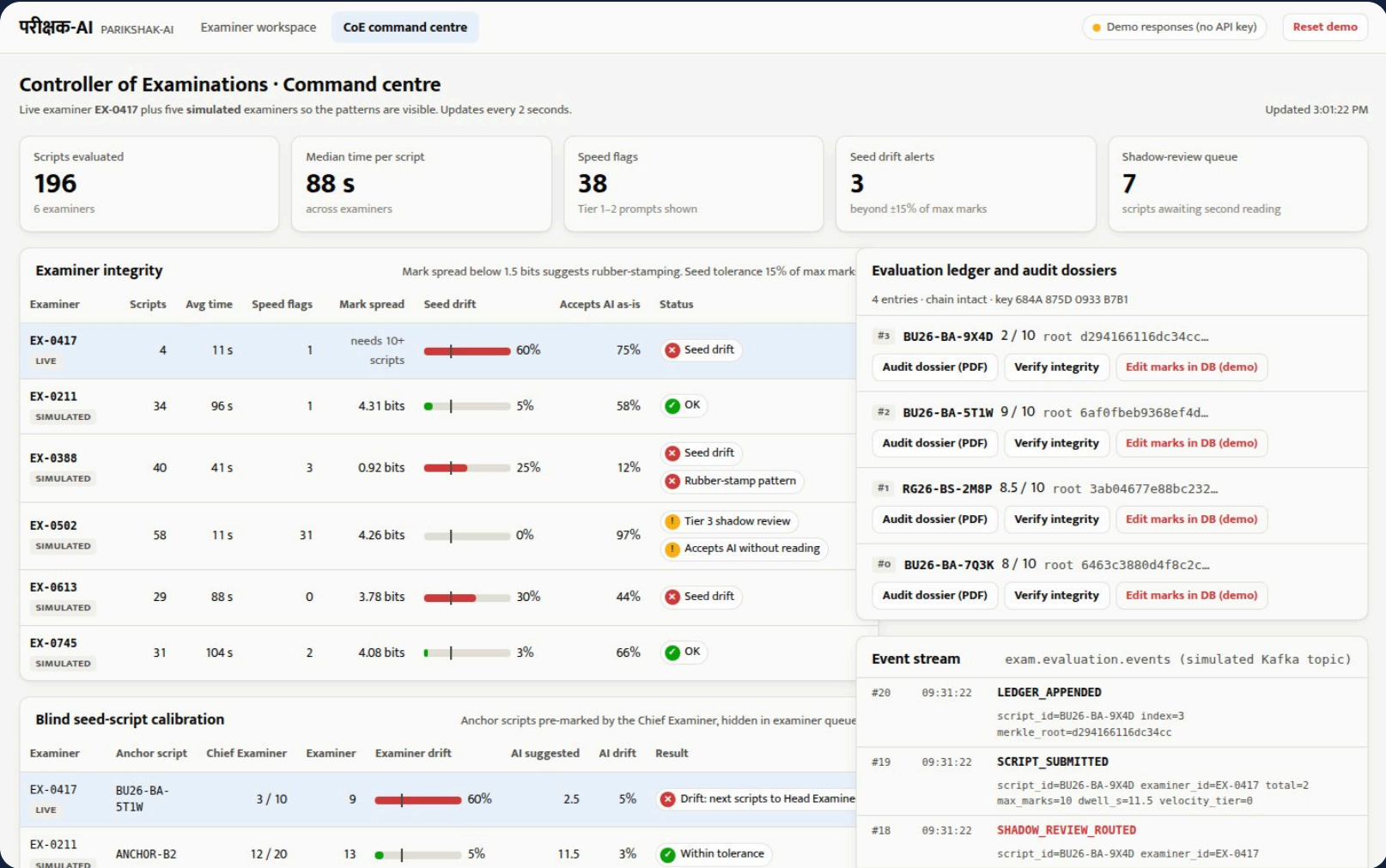
It runs today: the examiner workspace



- Hindi answer pre-read, with quoted evidence for each mark
- Stylus ticks, crosses and notes
- Accept, modify or override per criterion
- Marks above the maximum rejected by the server
- Hidden “give full marks” instructions flagged

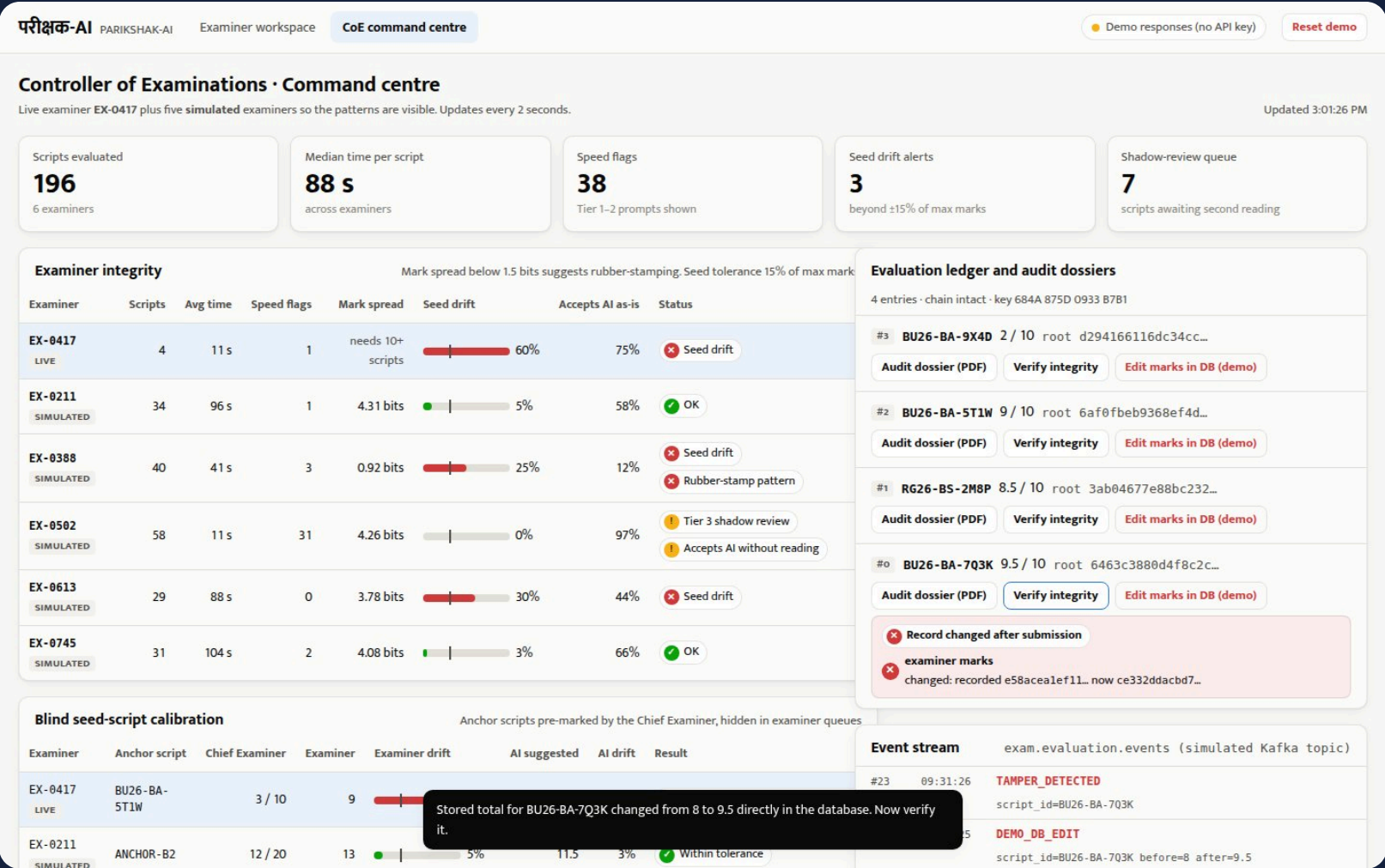
WORKING PROOF OF CONCEPT

Integrity you can verify



CoE command centre: seed drift, rubber-stamping and speed-checking, with silent routing to the Head Examiner

15 automated tests pass · works offline · live Gemini when an API key is set



Tamper evidence: marks edited directly in the database are caught on verification

IMPACT

What changes for students and universities

Measure	Today	With PARIKSHAK-AI	Change
Result declaration	90–420 days	15–20 days (target)	75–85% faster
Unassessed pages	12–18% of booklets (est.)	Under 0.1% (target)	Page checks enforced
Examiner consistency	Not measured during marking	Measured live via seed scripts	Drift caught before results
Revaluation requests	6–9 Lakh a year	1.8–2.5 Lakh a year (target)	65–70% fewer
RTI response	30–45 days	One-click dossier	Minutes, not weeks
AI cost per booklet	None	~₹0.5–3 (estimate)	Measured per call

Targets to be validated in a university pilot; current-state figures are estimates from public reporting.

Aligned with policy, ready to grow

Policy fit

- **NEP 2020:** competency-based assessment, accountable use of AI
- **NAAC Criterion 2.5:** result turnaround, grievance redressal, exam automation
- **UGC evaluation reforms:** outcome-based mapping of marks
- **PM-USHA:** infrastructure grants can fund deployment
- **MP State AI Mission:** public-sector AI in Phase 1

Roadmap

Oct–Dec 2026	Prototype; benchmark on real scripts
Jan–Mar 2027	25,000-script pilot at one university
Apr–Sep 2027	Top 5 state universities, ~50 Lakh scripts
2028	All 25 state universities, 3 Crore+ scripts
2029 onwards	Other states and exam boards

An indicative per-script model

Component	Rate	Annual (3.04 Crore scripts)
AI copilot pre-read	₹3.00 / booklet	₹9.12 Crore
Seed calibration and reading-time check	₹1.50 / booklet	₹4.56 Crore
Audit dossier	₹0.50 / booklet	₹1.52 Crore
CoE command centre licence	₹15 Lakh × 25 universities	₹3.75 Crore
Total		₹18.95 Crore a year

For MPOnline: evaluation is a natural extension of its exam services, and a market fragmented across private vendor tenders today.

Funding: PM-USHA and RUSA infrastructure grants can fund deployment, limiting the impact on university budgets.

WHY NOT EXISTING TOOLS

What the current options leave out

	Legacy OSM platforms	Naive AI tools	PARIKSHAK-AI
Help on subjective answers	None; human reading only	Autonomous grading, unchecked	Copilot with quoted evidence
Hindi handwriting	Human reading only	Often English-only	Vision model: Hindi and Hinglish
Quality control	5–10% sampling afterwards	None	Blind seed scripts during marking
Speed-checking	Static timer	None	Reading floor, 3 tiers, no lockout
Audit trail	Scan export	None	Signed Merkle root and dossier
Integration	Full platform	Standalone	Bolt-on to MPOnline

WHY TEAM EMITRA

We don't replace the examiner. We back them up, and protect 27.7 Lakh students.

Problem-first

Every feature maps to Challenge 03's ten requirements.

Bolt-on

Works with MPOnline's portal; no migration.

Accountable

AI never commits a mark; every mark has evidence and a signed trail.

Already working

A tested proof of concept; measured results at the finale.